MFCSP: CRYSTAL STRUCTURE PREDICTION USING MEAN FLOW GENERATIVE MODELS

Sourin Dey
Department of Computer Science and Engineering
University of South Carolina
Columbia, SC 29201

Dipannoy Gupta
Department of Computer Science and Engineering
University of South Carolina
Columbia, SC 29201

Lai Wei, Sadman Omee
Department of Computer Science and Engineering
University of South Carolina
Columbia, SC 29201

Jianjun Hu*
Department of Computer Science and Engineering
University of South Carolina
Columbia, SC 29201
jianjunh@cse.sc.edu

ABSTRACT

Crystal structure prediction (CSP) is a central task in computational materials discovery. Generative models such as CDVAE, DiffCSP, MatterGen, and CrystalFlow have made it possible to learn the distribution of stable crystals directly instead of searching the potential-energy surface. Inference remains the bottleneck: diffusion and flow-matching samplers integrate a generative trajectory over tens to thousands of sequential network evaluations per candidate. We introduce MFCSP, a CSP model built on MeanFlow, which learns the average rather than the instantaneous velocity of the probability-flow trajectory and can therefore generate a complete structure in as little as a single network evaluation. We also introduce pMF-CSP, a companion variant built on the latent-free pixel mean flow formulation. The velocity field is parameterized by a chemistry- and symmetry-aware Transformer conditioned on a canonical atom ordering, a global composition embedding, per-token chemistry features, and a coarse crystal-system token used only as an auxiliary training signal, so inference remains formula-only. On the MP-20 benchmark, generating 20 candidates per target, a single MeanFlow step matches CrystalFlow's match rate (78.38% vs. 78.34%) using $100\times$ fewer network evaluations and roughly an order of magnitude less wall-clock time; five steps raise this to 83.64%, above CrystalFlow (78.34% at 2,000 evaluations) and DiffCSP (77.93% at $\sim$20,000), still at $20\times$ fewer evaluations. MFCSP generates 10k structures in 0.39–1.31 min, compared with 6.5 min for CrystalFlow and 76.1 min for DiffCSP. On the stricter, formula-only CSPBench protocol, which withholds cell size and space group and scores symmetry recovery, MFCSP reaches $72/72/65\%$ structure, space-group, and consensus match rates at five sampling steps. CrystalFlow attains higher rates ($78/73/68$) at its full 100-step budget, but collapses to $49/32/31\%$ when held to the same five-step budget, less than half the space-group and consensus rates. MFCSP therefore improves accuracy per network evaluation rather than peak accuracy alone.

Keywords crystal structure prediction · materials discovery · benchmark · neural network potential · deep learning

1 Introduction

Crystal structure prediction (CSP) addresses the fundamental challenge of determining how atoms arrange themselves in periodic solids from a given chemical composition alone [1, 2]. The atomic arrangement is the state variable that links composition to behavior: once the structure of a compound is known, first-principles methods can predict its stability, electronic structure, ionic transport, catalytic activity, and mechanical response with quantitative accuracy [3, 4]. CSP is therefore the enabling step of *in silico* materials discovery, allowing hypothetical compounds, polymorphs, and pressure-stabilized phases to be assessed before they are ever synthesized [5]. Structure searches have in this way delivered materials that lie beyond chemical intuition, from high-temperature superconducting superhydrides to

metastable battery cathodes and functional photovoltaic and thermoelectric phases [6, 3]. Getting the structure right also means getting its symmetry right. The space group and Wyckoff positions of a crystal are physical constraints: they fix site multiplicities and local environments, dictate which property tensors and electronic band degeneracies are allowed, and underpin the symmetry indicators of electronic topology [7]. A prediction that is geometrically close but crystallographically wrong can imply the wrong physics, so faithful recovery of symmetry is a central requirement for any CSP method [8].

Historically, CSP was formulated as a global optimization problem on a potential-energy surface. Candidate atomic arrangements and lattice parameters were sampled over a large configurational space and then relaxed or ranked using first-principles energy calculations [9, 1, 10, 11]. Ab initio random structure searching samples configurations stochastically before relaxing them with density functional theory (DFT) [12]; evolutionary algorithms such as USPEX evolve populations toward low-energy minima [13]; and particle-swarm optimization, as implemented in CALYPSO, explores the energy landscape using swarm intelligence [14]. These methods are physically rigorous and remain important tools in high-pressure and exploratory materials science, but their reliance on repeated DFT relaxations makes them computationally expensive. The cost becomes particularly severe as the number of atoms, unit-cell size, and chemical search space increase. Moreover, many polymorphs can have similar energies, so successful prediction requires both efficient exploration and accurate energy ranking.

The advent of deep generative modeling reframed CSP as learning the distribution of stable crystal structures from materials databases, thereby amortizing much of the search cost into a trained model. The Crystal Diffusion Variational Autoencoder (CDVAE) pioneered this direction by combining a variational autoencoder with a diffusion-based decoder that denoises crystal representations toward plausible, low-energy configurations [15]. DiffCSP subsequently formulated CSP as a joint equivariant diffusion process over fractional coordinates and lattice parameters, using periodic E(3)-equivariant modeling to respect the symmetries of crystals [16]. DiffCSP++ incorporated explicit space-group and Wyckoff-position constraints into the generative process, further narrowing the search space to symmetry-compatible structures [17].

Crystallographic symmetry has also become an explicit design principle in generative CSP. CrystalFormer uses a space-group-informed autoregressive Transformer that generates symmetry-inequivalent atoms through Wyckoff positions, reducing the effective complexity of crystal generation [18]. SymmCD instead decomposes a crystal into its asymmetric unit and the symmetry transformations needed to reconstruct the full structure, allowing diffusion to operate in a compact symmetry-aware representation [19]. WyckoffDiff uses a discrete diffusion model over space groups and Wyckoff-position occupancies, enforcing symmetry by construction rather than predicting unconstrained atomic coordinates [20]. Beyond unconditional or composition-conditioned CSP, MatterGen demonstrated property-conditioned diffusion for steering crystal generation toward target chemistries and properties across the periodic table [21], while autoregressive language models such as CrystaLLM showed that crystals can also be generated directly as text by modeling the Crystallographic Information File (CIF) format [22].

Flow-based models provide an alternative to diffusion for learning distributions over periodic materials. FlowMM introduced Riemannian flow matching for crystal generation, modeling lattice parameters, periodic fractional coordinates, and atom types on manifolds that reflect crystal-specific geometry and symmetries [23]. CrystalFlow combined conditional flow matching with a graph-based equivariant network and symmetry-aware crystal representations to generate lattice parameters, atomic coordinates, and compositions, while requiring substantially fewer integration steps than diffusion-based models [24]. These approaches improve sampling efficiency, but they still generally require numerical integration over multiple network evaluations.

A common thread runs through all of these families, however: generation remains a sequential process. Diffusion models denoise over hundreds to thousands of steps, flow-matching models integrate a velocity field over many evaluations, and autoregressive models emit a structure token by token, so richer symmetry priors and conditioning improve quality without removing the per-candidate inference cost. Reducing that cost to a single or a handful of network evaluations is what motivates one-step generative modeling for CSP. MeanFlow characterizes a generative trajectory through an average velocity over a time interval rather than an instantaneous velocity, enabling one-step generation without distillation or pre-training [25]. Pixel MeanFlow extends this idea to latent-free generation by predicting the denoised signal directly in raw pixel space and deriving the corresponding flow field [26]. Bringing one-step formulations to periodic crystal representations is the central focus of this work.

Evaluating such a model calls for two complementary tests, and we adopt them in sequence. We first place MFCSP on the standard MP-20 benchmark [15, 16], the common proving ground on which diffusion and flow-matching models such as DiffCSP, FlowMM, and CrystalFlow report their any-of-k match rates. Evaluating under this identical protocol lets us compare MFCSP head-to-head with those baselines and answer a concrete question: can a one- or few-step generator match the accuracy of samplers that integrate hundreds to thousands of network evaluations? The match rate alone, however, rewards geometric proximity and can be inflated by duplicate- or polymorph-contaminated splits, so

a high score does not by itself show that a model has learned the underlying crystallography [8]. We therefore turn to CSPBench [27], a curated suite of 180 held-out structures that span binary to quaternary compositions and graded difficulty, are presented formula-only (without the number of formula units in the cell or any space-group label) and are scored not only by structure matching but also by space-group and consensus match rates. Because CSPBench withholds symmetry information at inference and credits a prediction only when structure and space group agree simultaneously, it directly probes whether a model recovers the correct symmetry, which is what our symmetry-aware conditioning targets. This two-stage evaluation (efficiency and accuracy on MP-20, symmetry fidelity on CSPBench) frames the results reported below.

This work makes three contributions. First, we adapt one-step MeanFlow generation to crystal structure prediction, showing that a complete periodic structure can be produced in a single network evaluation rather than by integrating a generative trajectory. Our method achieves a higher match rate than multi-step diffusion and flow-matching baselines at 20 to 200 times fewer network function evaluations and 5 to 58 times lower measured wall-clock time. Second, we introduce a chemistry- and symmetry-aware Transformer parameterization of the average-velocity field, and show by ablation that canonical atom ordering, a global composition embedding, per-token chemistry features, and a coarse crystal-system auxiliary token contribute additively; the symmetry token improves space-group agreement while remaining unused at inference. Third, we show that the recipe is not specific to one objective: pMF-CSP, the same model trained with the latent-free pixel mean flow formulation [26], reaches 83.56 under the identical budget, within a tenth of a point of MFCSP. This suggests that the conditioning and representation, beyond the particular one-step objective, drive most of the performance.

2 Method

2.1 Problem Setting and Representation

We consider crystal structure prediction in two related settings. In the formula-only setting, the model receives a chemical composition and generates a periodic crystal without access to the target space group at inference. In the full-composition setting used by the standard MP-20 protocol, the atom multiset in the primitive cell is supplied as the composition condition. A crystal containing N atoms is represented in raw, all-atom form by its atomic numbers $\{z_i\}_{i=1}^{N}$, Cartesian coordinates $\mathbf{X} \in \mathbb{R}^{N\times 3}$, and lattice matrix $\mathbf{L} \in \mathbb{R}^{3\times 3}$. The lattice matrix is flattened into nine continuous parameters. Atomic numbers are fixed by the composition condition, while the model predicts the continuous coordinates and lattice parameters.

This representation avoids a latent autoencoder and preserves the explicit periodic structure needed for downstream symmetry analysis and structure matching. The distinctions between the two composition conditions, including whether the cell composition and cell size are provided, are specified in the evaluation protocols below.

2.2 Few-Step Generative Backbone

Our generator is a few-step MeanFlow model [25]. Standard flow-matching methods learn the instantaneous velocity of a probability-flow trajectory and therefore require numerical integration over many sequential network evaluations. MeanFlow instead learns the average velocity between two times, r and t, allowing each update to cover a relatively large interval. Given a state $\mathbf{x}_t$ at time t, the predicted mean velocity $u_\theta(\mathbf{x}_t, t, r)$ produces the state at r as

$$\widehat{\mathbf{x}}_r = \mathbf{x}_t - (t-r)u_\theta(\mathbf{x}_t, t, r).$$

We apply this update over a five-step schedule in the reported experiments. The mean-velocity field is trained with the MeanFlow self-consistency objective, whose Jacobian–vector products are evaluated using forward-mode automatic differentiation. Classifier-free condition dropout is used during training, and the corresponding guidance formulation is used during sampling.

2.3 Pixel MeanFlow Variant

To test whether the results depend on the particular MeanFlow parameterization, we also train pMF-CSP with the latent-free Pixel MeanFlow formulation [26]. Standard MeanFlow predicts the average velocity directly. Pixel MeanFlow instead predicts a denoised version of the input state, $\widehat{\mathbf{x}}_\theta$, and converts this prediction into an average-velocity estimate through

$$\widehat{\mathbf{x}}_\theta = f_\theta(\mathbf{x}_t, r, t), \qquad \widehat{\mathbf{u}}_\theta(\mathbf{x}_t, r, t) = \frac{\mathbf{x}_t - \widehat{\mathbf{x}}_\theta}{t}.$$

The resulting velocity field is used in the Pixel MeanFlow velocity-space self-consistency objective, with the required Jacobian–vector products computed by forward-mode automatic differentiation. Here, "pixel" denotes latent-free

direct modeling: the same parameterization is applied to the raw crystal-state vector rather than to an image or a pretrained latent representation. The crystal representation, Transformer architecture, conditioning signals, training data, and five-step sampling budget are unchanged between MFCSP and pMF-CSP; only the output parameterization and associated training objective differ.

2.4 Chemistry- and Symmetry-Aware Transformer

The velocity field is parameterized by a sequence Transformer that treats each atom as a token. Chemical and symmetry information enters through atom ordering, per-token features, and global conditioning.

Atom ordering. Because self-attention is sensitive to token order, atoms are presented in a canonical ordering inspired by MCFlow [28]. During preprocessing, atoms are sorted by Pauling electronegativity. For symmetry-aware training examples, atoms are additionally grouped into space-group orbits and ordered by Wyckoff letter. This symmetry analysis is performed once with PYMATGEN [29] and its SPGLIB backend [30]. The space-group-dependent information is used only during training or auxiliary preprocessing; the formula-only inference branch does not receive a space-group label.

Per-token features. Each atom token combines a learned atom-type embedding with fixed chemistry descriptors: Pauling electronegativity, atomic radius, period, and group. These descriptors are obtained from PYMATGEN [29]. The token representation also contains the atom's current coordinates and the shared lattice parameters.

Global conditioning. A conditioning vector modulates every Transformer block through adaptive layer normalization:

$$\mathbf{c} = \mathbf{e}_t + \mathbf{e}_r + \mathbf{e}_{\text{formula}} + \mathbf{e}_{\text{sys}},$$

where $\mathbf{e}_t$ and $\mathbf{e}_r$ are embeddings of the two MeanFlow times, $\mathbf{e}_{\text{formula}}$ is a global composition embedding computed from normalized element counts, and $\mathbf{e}_{\text{sys}}$ is a coarse crystal-system embedding. The crystal system is obtained by mapping the space-group number to one of seven classes: triclinic, monoclinic, orthorhombic, tetragonal, trigonal, hexagonal, and cubic. We deliberately use this seven-way label rather than a 230-way space-group embedding. Each crystal-system class is shared by many space groups and therefore receives substantially more training examples, providing a broad symmetry prior without requiring the model to commit to an exact space group.

The crystal-system embedding is an auxiliary training signal rather than an inference condition. Following classifier-free guidance, it is replaced by a null token for a fraction of training examples, so the shared network learns both symmetry-conditioned and unconditioned velocity fields. At formula-only inference, the crystal-system input is always null and the model uses its formula-only branch. The resulting improvement is therefore interpreted as a multi-task regularization effect: learning the auxiliary symmetry-conditioned task improves the representation used by the unconditioned generator. All inference-time conditioning signals are derivable from the composition; the space group is not required at generation.

2.5 Implementation and Training

We train on the MP-20 subset of the Materials Project [31, 15], which contains structures with at most 20 atoms per unit cell, using the fixed DiffCSP split. The 180 CSPBench evaluation structures are removed from the training set, so no evaluation structure is seen during training. The denoiser is a 12-layer Transformer with hidden size 768 and 12 attention heads; atom types use a 32-dimensional learned embedding. Models are implemented in PyTorch [32] and trained for 700 epochs in mixed `bf16` precision with AdamW at a constant learning rate of 10^{-4} and batch size 256 on a single NVIDIA A100 GPU. The MeanFlow flow ratio is 0.25, the classifier-free-guidance dropout probability is 0.8, and the guidance scale is 1.5. The Jacobian–vector products in the MeanFlow objective are computed with forward-mode automatic differentiation.

Training-data scope. We evaluate under two protocols: the standard MP-20 benchmark [15, 16], which gives the full unit-cell composition and scores raw candidates by Match Rate and RMSE (Protocol B), and CSPBench [27], which gives only the formula and scores an energy-ranked top-5 shortlist by structure, space-group, and consensus rates (Protocol A). We train a separate model for each, in both cases on MP-20 with the evaluation structures held out: the 180 CSPBench targets for Protocol A, and the standard test split for Protocol B. We keep training to MP-20 for three reasons.

First, all generative baselines we compare against (CDVAE [15], DiffCSP [16], FlowMM [23], CrystalFlow [24]) are trained on the same MP-20 split. Fixing the training data lets us attribute our gains to the one-step objective rather than to more data, and keeps our numbers comparable to published results.

Second, for the TCSP 2.0 [33] comparison specifically: TCSP 2.0 has no trained model and no MP-20 split. It retrieves and substitutes from a 731,293-entry library spanning the Materials Project [31], Materials Cloud [34], C2DB [35], and GNoME [36]. Since the CSPBench targets come from the Materials Project, training our model on MP-20 keeps it on the same pool TCSP 2.0 draws from. Adding the larger library would let our model use sources the comparison is meant to hold fixed.

Third, more data does not help a generative model the way it helps a template method. A template library is a lookup index, so wider coverage always helps. A generative model instead learns a distribution, and the extra sources are far from real bulk crystals: GNoME [36] is mostly DFT-hypothetical, with only 736 of 2.2 million entries experimentally confirmed, and C2DB [35] is two-dimensional. We leave training on larger, carefully cleaned data to future work.

2.6 Evaluation Protocols

We evaluate the model under two established but mutually incompatible protocols. They differ in the composition supplied to the model, candidate generation and selection, structural relaxation, and matching criterion. Their numerical results are therefore reported in separate tables and are not compared directly.

Protocol A: CSPBench [27]. CSPBench contains 180 held-out structures and evaluates an energy-ranked top-5 shortlist. The model receives only the reduced chemical formula; the number of formula units in the cell is not supplied, and no space-group label is provided. For each formula, we generate 50 candidate structures using the five-evaluation sampling configuration used for this protocol. Each candidate is locally relaxed with the ORB-v3 universal machine-learned interatomic potential [37]. The relaxed candidates are ranked by CHGNet-predicted energy per atom [38], and the five lowest-energy candidates form the top-5 shortlist. The reference structure is used only for scoring and never for candidate selection, so this is a genuine top-5 metric rather than recall over all 50 generated candidates.

A candidate is counted as a structural match if `pymatgen`'s `StructureMatcher` [29] accepts it against the reference using $\mathrm{ltol} = 0.2$, $\mathrm{stol} = 0.3$, and $\mathrm{angle_tol} = 5^\circ$ on primitive cells. Both candidate and reference structures are lightly symmetry-refined at $\mathrm{symprec} = 0.1\,\text{Å}$ before matching, so the structural test is independent of the space-group assignment. A candidate is counted as a space-group match when the integer label from 1 to 230 returned by `SpacegroupAnalyzer` with its `spglib` backend [30], evaluated at the same tolerance, equals the label assigned to the reference. The same procedure is applied to generated and reference structures, making the assignment deterministic and symmetric.

The structure-match rate and space-group-match rate are the fractions of the 180 formulas for which at least one top-5 candidate matches the reference under the corresponding criterion. The consensus rate is the fraction of formulas for which one top-5 candidate matches in both structure and space group simultaneously. Formulas for which no valid candidate is produced are counted as failures.

Protocol B: CDVAE, DiffCSP, and CrystalFlow. This protocol follows the raw, any-of-k MP-20 evaluation used by CDVAE [15], DiffCSP [16], and CrystalFlow [24]. The model receives the full unit-cell composition, namely the atom multiset of the primitive cell, so the number of atoms is known. For each structure in the 9,046-structure MP-20 test split, we generate k candidates without relaxation or energy-based selection. Each reference structure is scored only against its own k generated candidates.

A structure is counted as matched if `StructureMatcher.get_rms_dist` returns a value within $\mathrm{ltol} = 0.3$, $\mathrm{stol} = 0.5$, and $\mathrm{angle_tol} = 10^\circ$ for at least one candidate. The match rate is the fraction of the 9,046 test structures with at least one such match. RMSE is computed from the minimum normalized root-mean-square displacement among the candidates for each matched structure and then averaged over matched structures.

Protocol distinction. Protocol A withholds the cell size, relaxes and energy-ranks candidates, and uses the CSPBench matching criterion. Protocol B supplies the full cell composition, evaluates raw generated structures, performs no energy-based selection, and uses the MP-20 matching criterion. Because these choices alter both the information available to the model and the scoring procedure, the absolute rates from the two protocols measure different tasks and are not directly comparable. The CSPBench set is also dominated by relatively small cells, for which a single formula unit is common; consequently, withholding the cell size has a different effect there than it does in the full MP-20 protocol.

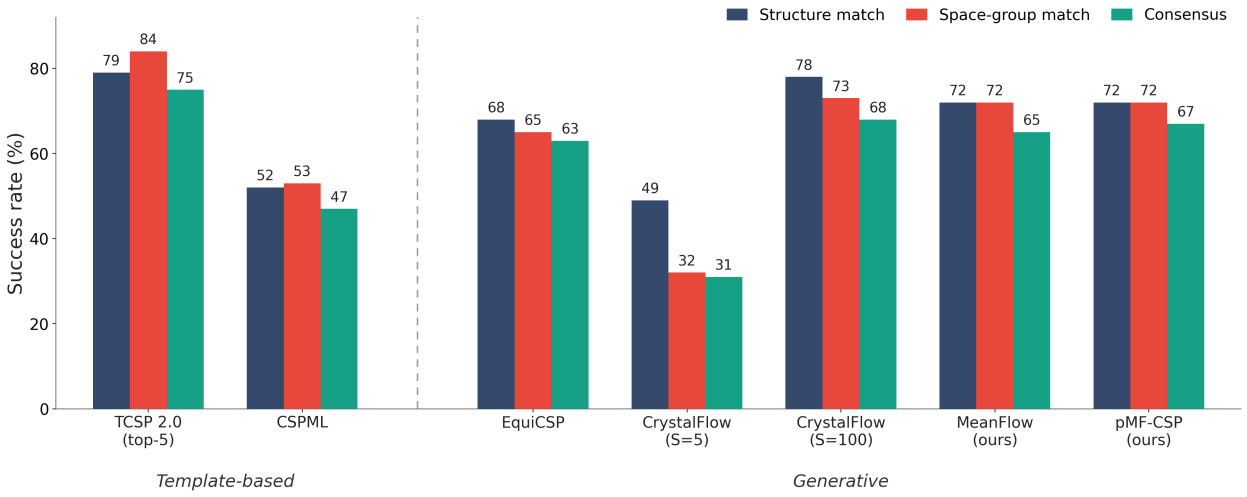


Figure 1: Success rates for StructureMatcher, space-group match, and consensus on the 180-formula CSPBenchmark test set, grouped by method family. A prediction counts toward the consensus rate only when the structure match and the space-group match both succeed. All entries use formula-only conditioning and top-5 selection under the evaluation protocol of protocolA. The two families are not matched on data access: template-based methods retrieve and substitute from large structure libraries (TCSP 2.0 draws on 731,293 templates spanning the Materials Project, Materials Cloud, C2DB and GNoME [33]), whereas the generative models are trained on MP-20 alone, and are further separated by inference budget: MeanFlow and pMF-CSP use $S = 5$ integration steps, and CrystalFlow is shown at both $S = 5$ (the matched few-step budget) and $S = 100$ (its full budget, $20\times$ more network evaluations). At its full budget CrystalFlow attains the highest generative rates ($78/73/68$), while the five-step MeanFlow and pMF-CSP follow closely at $72/72/65$ and $72/72/67$; held to the same five-step budget CrystalFlow collapses to $49/32/31$, so at equal cost the MeanFlow models lead decisively, especially on symmetry, and reach CrystalFlow-competitive accuracy at one-twentieth of the network evaluations. Values for TCSP 2.0, CSPML and EquiCSP are as published [33]; CrystalFlow, MeanFlow and pMF-CSP were measured here.

Table 1: Top-k performance comparison of MeanFlow, Pixel MeanFlow, and Crystal Flow on the 180-formula CSPBench split. MeanFlow and Pixel MeanFlow sample with $S = 5$ integration steps; Crystal Flow is reported at both $S = 5$ (the matched few-step budget) and $S = 100$ (its full multi-step budget, $20\times$ more network evaluations). At the matched five-step budget Crystal Flow collapses on all three metrics, especially symmetry, and only overtakes the MeanFlow models when its step budget is raised twenty-fold. All rows: $k = 50$ samples per formula, ORB relaxation, CHGNet energy ranking, StructureMatcher $(\mathrm{ltol}, \mathrm{stol}, \mathrm{angle}) = (0.2, 0.3, 5)$, space groups from `SpacegroupAnalyzer` at $\mathrm{symprec} = 0.1$.

Model	**Top-1**			**Top-5**			**Top-10**		
	SMR	**SGMR**	**CR**	**SMR**	**SGMR**	**CR**	**SMR**	**SGMR**	**CR**
MeanFlow (S=5)	68.89	64.44	58.33	71.67	71.67	65.00	74.44	73.33	66.67
Pixel MeanFlow (S=5)	67.22	64.44	58.89	71.11	71.67	66.67	72.22	73.33	68.33
Crystal Flow (S=5)	46.67	30.00	28.89	49.44	32.22	31.11	52.78	35.00	33.89
Crystal Flow (S=100)	75.00	64.44	58.89	77.78	72.78	67.78	80.00	76.67	72.22

3 Results & Discussion

Component ablation. Figure 2 shows that the four conditioning signals contribute additively. From a raw all-atom MeanFlow with no chemical or symmetry priors ($58/47/44$ for structure / space-group / consensus match), the MCFlow-style canonical atom ordering gives the single largest jump in symmetry recovery ($+14$ space group and $+12$ consensus, with $+3$ structure): supplying the order-sensitive Transformer with a deterministic, symmetry-aware token sequence removes a major source of nuisance variation. The global formula embedding contributes the largest structure-match increment ($+8$ structure, with $+5$ space group and $+3$ consensus), the per-token chemistry features add a further gain ($+2/+1/+3$), and the coarse crystal-system auxiliary token completes the model at $72/72/65$. Consistent with its

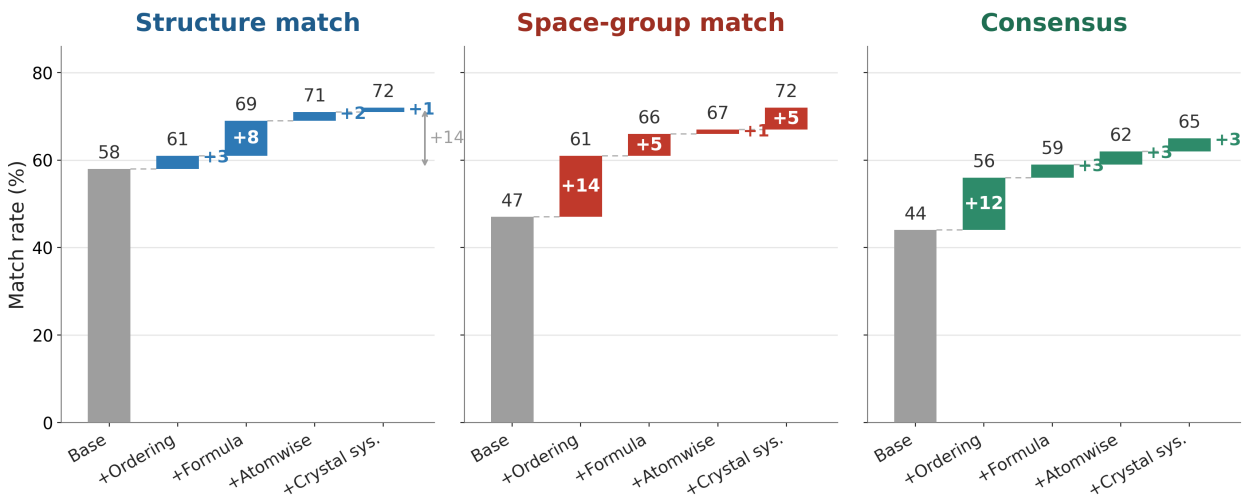


Figure 2: Additive effect of the four conditioning components on MeanFlow crystal-structure prediction, expanding the ablation reported in Supplementary Table S3. Each panel is a separate metric; the grey bar is the raw all-atom baseline (no chemical or symmetry priors) and each coloured block is the increment contributed by switching on one component, in the order they are added. Canonical atom ordering gives the single largest jump in space-group match ($+14$) and consensus ($+12$), the global formula embedding contributes the largest structure-match increment ($+8$), the per-token chemistry features add a further gain, and the coarse crystal-system auxiliary token (null at inference) adds a final $+5$ to space-group match while still improving structure and consensus, completing the model at $72/72/65$. All rows use an identical five-step sampling budget, so the gains reflect conditioning rather than extra integration steps.

role as a symmetry regularizer (Section 2.4), this token—though replaced by a null token at inference—adds a final $+5$ to space-group match while still improving structure and consensus. All rows use an identical five-step sampling budget, so the gains reflect conditioning rather than extra integration steps.

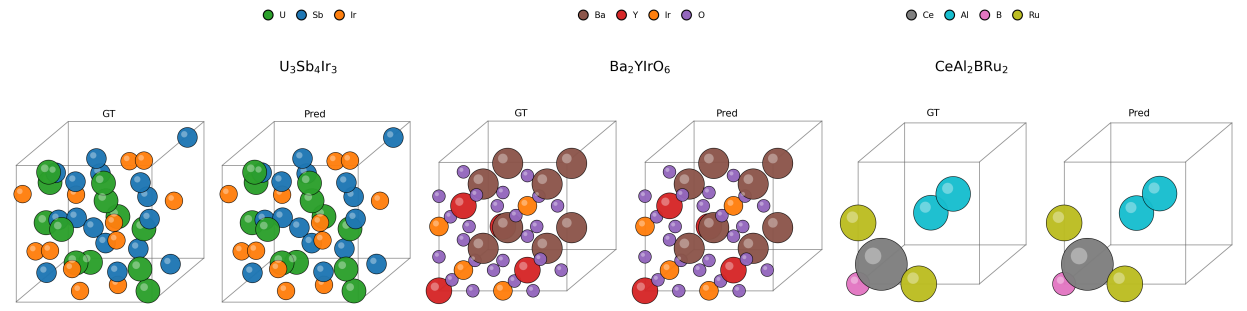


Figure 3: Ground truth (GT) versus predicted (Pred) crystal structures for three multi-element consensus examples: (a) $U_3Sb_4Ir_3$, (b) Ba_2YIrO_6, and (c) $CeAl_2BRu_2$. All three achieve both StructureMatcher and space-group agreement with the ground truth. Predicted structures are shown in the ground truth's coordinate frame (translation/axis alignment only, via pymatgen's `StructureMatcher.get_s2_like_s1`, the same transformation used internally to establish the match) for direct visual comparison; reported match statistics are computed on the unaligned structures.

Velocity-field parameterization. Table 2 isolates the effect of the average-velocity predictor with the selection protocol held fixed. Among the average-velocity predictors we tried, the chemistry- and symmetry-aware Transformer and the latent-free Pixel variant are the strongest and perform comparably ($72/72/65$ and $71/72/67$), both clearly ahead of an equivariant GNN backbone ($70/64/60$); this near-parity of the Transformer and Pixel predictors mirrors our broader finding that the conditioning and representation, rather than the particular one-step objective, drive accuracy. The informative comparison to CrystalFlow, a strong flow-matching baseline, is at matched compute. Held to the

Table 2: Comparison of average-velocity predictors in the MeanFlow model against the CrystalFlow baseline. The Transformer and Pixel predictors use the best configuration from the ablation in Figure 2 (atom **Ord**ering + **Form**ula embedding + per-token **Atom**wise chemistry features + coarse crystal-**sys**tem auxiliary conditioning). All entries use the top-5, CHGNet-ranker selection protocol; results are percentages. The MeanFlow predictors use $S = 5$ integration steps, whereas the CrystalFlow baseline uses its full $S = 100$ budget ($20\times$ more network evaluations); at a matched $S = 5$ budget CrystalFlow scores $49/32/31$ (Table 1). Best in **bold**.

Metric	**Average Velocity Predictor**			**CrystalFlow**
	Transformer (Ord+Form+Atom+Sys)	**Equivariant GNN**	**Pixel (Ord+Form+Atom+Sys)**	
Structure Match Rate	72	70	71	**78**
Space Group Match Rate	72	64	72	**73**
Consensus Rate	65	60	67	**68**

same five-step budget, CrystalFlow reaches only $49/32/31$ on structure, space-group, and consensus (Table 1), less than half the MeanFlow model's space-group and consensus rates. Standard flow matching degrades sharply when the probability-flow trajectory is integrated in only a handful of steps, whereas the average-velocity formulation does not; recovering symmetry at low step count is exactly where the MeanFlow parameterization pays off. CrystalFlow overtakes our models only when its integration budget is raised twenty-fold to $S = 100$ ($78/73/68$), and even then its lead narrows to near-parity on symmetry (73 vs. 72) while costing twenty times the network evaluations. The MeanFlow predictors thus trade a small margin of peak accuracy for a large reduction in inference cost.

Table 3: Crystal structure prediction on the **MP-20** benchmark. k = candidates per target, S = integration steps, $\mathrm{NFE} = S \times k$ = network evaluations per target, t = wall-time to generate 10k structures. MFCSP is the best of four seeded training runs of the identical configuration, evaluated at its best-validation checkpoint (seed 11, epoch 699); pMF-CSP is a single training run at its final checkpoint. Both score each target against only the k candidates generated for it ($n = 9046$). Baseline numbers are as published. Baseline t measured on NVIDIA A800 [24], ours on NVIDIA A100 (same compute class), batch $= k = 20$; t depends on S only, and the two methods are equal in wall-clock within measurement spread. Best per column in **bold**. At $k = 20$, MFCSP reaches 83.64% and pMF-CSP 83.56%, both above CrystalFlow at $k = 100$ (82.49) with $100\times$ fewer network evaluations.

Method	k	S	NFE	MR (%) ↑	RMSE ↓	t (min/10k) ↓
CDVAE [15]	1	∼5,000	∼5,000	33.90	0.1045	–
DiffCSP [16]	1	1,000	1,000	51.49	0.0631	76.1
FlowMM [23]	1	50–100	∼100	61.39	0.0566	13.2
CrystalFlow [24]	1	100	100	62.02	0.0710	6.5
pMF-CSP (ours)	1	5	**5**	53.74	0.0564	1.29
DiffCSP [16]	20	1,000	20,000	77.93	**0.0492**	76.1
CrystalFlow [24]	20	100	2,000	78.34	0.0577	6.5
MFCSP (ours)	20	1	20	78.38	0.0783	**0.39**
MFCSP (ours)	20	5	100	**83.64**	0.0619	1.31
pMF-CSP (ours)	20	5	100	83.56	0.0566	1.29
CrystalFlow [24]	100	100	10,000	82.49	0.0513	6.5

Accuracy versus inference cost on MP-20. Table 3 evaluates MFCSP and pMF-CSP on the standard MP-20 split using the DiffCSP known-Z, any-of-k protocol without structural relaxation. This evaluation differs from the formula-only, relaxed top-5 protocol used in Figure 2 and Table 2; the two sets of results therefore measure different tasks and should not be compared directly.

The main advantage of MeanFlow is its lower inference cost. At $k = 20$, MFCSP achieves a match rate of 78.38% with one MeanFlow step and 83.64% with five steps, while pMF-CSP reaches 83.56% with five steps. The corresponding numbers of network function evaluations are only 20 and 100 per target, respectively. By comparison, CrystalFlow requires $2{,}000$ evaluations at $k = 20$, and DiffCSP requires approximately $20{,}000$. Thus, the five-step MFCSP and pMF-CSP results use 20–$200\times$ fewer network evaluations while achieving higher match rates than these baselines.

At the same 100-evaluation budget, MFCSP reaches 83.64% and pMF-CSP reaches 83.56%, compared with 62.02% for CrystalFlow using one candidate and 100 integration steps. The MeanFlow models therefore use the budget to generate multiple coarse candidates, whereas CrystalFlow uses it to integrate one candidate more finely.

The five-step refinement improves MFCSP from 78.38% to 83.64% in match rate and reduces RMSE from 0.0783 to 0.0619. pMF-CSP obtains an RMSE of 0.0566, comparable to FlowMM (0.0566) and CrystalFlow (0.0577), although DiffCSP remains lower at 0.0492. The generation times reported in Table 3 are 0.39 minutes for MFCSP with one step, 1.31 minutes for MFCSP with five steps, 1.29 minutes for pMF-CSP, 6.5 minutes for CrystalFlow, and 76.1 minutes for DiffCSP when generating 10,000 structures.

Overall, a five-step MeanFlow schedule substantially reduces network evaluations while retaining competitive structural accuracy and RMSE.

Relaxation behavior under a machine-learned potential. We next examine how much local relaxation the generated structures require, using SiO_2 as a case study following CrystalFlow [24]. We stress at the outset that this analysis is a *machine-learned-potential proxy*, not a first-principles measurement: all structures are relaxed with the ORB-v3 universal interatomic potential [37], and every energy, ionic-step count, and convergence rate reported in Figure 4 is defined by that potential and our optimizer settings rather than by DFT. The numbers are therefore internally consistent across the three generators, which are relaxed under an identical ORB-v3 protocol, but are not directly comparable to the DFT/VASP quantities reported in the original CrystalFlow SiO_2 study, and they inherit any systematic bias of the underlying potential. We use this diagnostic only as a cheap, relative measure of how close each generator's raw output sits to a relaxed local minimum, and not as a physical stability claim. Read this way, Figure 4 shows a clear and consistent trend. The per-structure energy drop on relaxation (panel a) has a median of 2.41 eV/atom for one-step MeanFlow (S=1), which falls to 0.34 eV/atom at five steps (S=5); the five-step generator is thus comparable to the CrystalFlow S=100 baseline (0.51 eV/atom) on this proxy while using 20× fewer network evaluations. The relaxation trajectories (panel b) tell the same story: S=1 structures need the most ionic steps and converge least often under ORB-v3, whereas S=5 and CrystalFlow S=100 converge faster and more completely. After relaxation, all three generators collapse onto the same low-energy peak (panel c), indicating that the samplers differ mainly in how far their *raw* outputs sit from the relaxed basin rather than in the basin they ultimately reach. This is consistent with the accuracy-cost picture above: additional MeanFlow steps sharply improve raw-structure quality, and five steps already yield structures that relax about as easily as those from a much longer CrystalFlow integration.

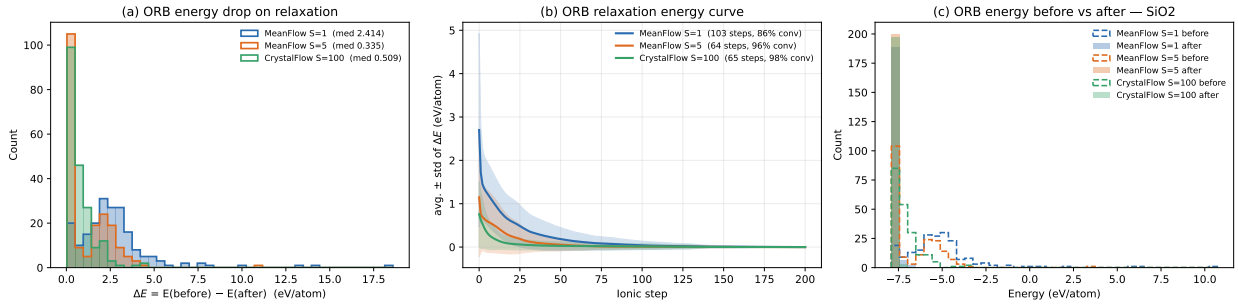


Figure 4: **ORB-v3 relaxation behavior for SiO_2 structures (machine-learned-potential proxy).** Structures generated by one-step MeanFlow (S=1), five-step MeanFlow (S=5), and CrystalFlow (S=100) are relaxed under an identical ORB-v3 protocol [37]. All quantities are defined by the potential and optimizer, *not* by DFT, and are not directly comparable to the DFT/VASP values in the original CrystalFlow study. **(a)** Per-structure energy drop on relaxation $\Delta E = E(\text{before}) - E(\text{after})$; legend reports the median ΔE. Smaller values indicate raw outputs already near a relaxed local minimum. **(b)** Average (± std) ΔE versus ionic step during relaxation; legend reports the mean ionic-step count and ORB-defined convergence rate. **(c)** Energy distributions before (dashed) and after (filled) relaxation; the three generators converge to the same low-energy peak. Five-step MeanFlow is competitive with CrystalFlow S=100 on this proxy at 20× fewer network evaluations.

4 Conclusion

We introduced MFCSP, a crystal structure prediction model built on the one-step MeanFlow formulation, together with pMF-CSP, a latent-free Pixel MeanFlow variant. By learning the average rather than the instantaneous velocity of the

probability-flow trajectory, both models generate a complete periodic structure in a handful of network evaluations, removing the sequential integration bottleneck that limits diffusion and flow-matching samplers. The average-velocity field is parameterized by a chemistry- and symmetry-aware Transformer conditioned on a canonical atom ordering, a global composition embedding, per-token chemistry descriptors, and a coarse crystal-system auxiliary token that improves symmetry recovery while being replaced by a null token at inference, so generation remains formula-only. On the MP-20 benchmark (Protocol B), MFCSP reaches a 83.64% match rate and pMF-CSP 83.56% at $k = 20$ with only 100 network evaluations per target, surpassing CrystalFlow (78.34% at 2,000 evaluations) and DiffCSP (77.93% at $\sim$20,000) while cutting wall-clock generation time by 5 to 58 times. On the 180-formula CSPBench suite (Protocol A), the model attains 72/72/65% structure, space-group, and consensus match rates at only five sampling steps; a CrystalFlow baseline held to the same budget reaches just 49/32/31%, and exceeds our model only when its integration budget is raised twenty-fold, showing that the average-velocity formulation gains accuracy per network evaluation rather than at peak accuracy. Ablations show that the four conditioning signals contribute additively, and the near-identical performance of the MeanFlow and Pixel MeanFlow objectives points to the conditioning and representation, rather than the particular one-step objective, as what drives accuracy. Together these results place CSP at a new point on the accuracy-cost frontier, demonstrating that high-quality crystal structures can be generated in a single or few network evaluations without sacrificing structural or symmetry fidelity.

Finally, our best-performing configuration adopts a deliberately simple geometric representation, and we report a negative result on the more crystallographically natural alternative. The MeanFlow model transports Cartesian positions and a raw nine-degree-of-freedom 3×3 lattice along Euclidean paths from a Gaussian prior, ignoring both the periodicity of fractional coordinates and the over-completeness of the unreduced cell. We implemented and trained the corresponding periodic variant—wrapped minimum-image geodesics on the torus $[0, 1)^3$ from a uniform prior, a six-parameter Niggli-reduced cell, periodic Fourier coordinate features, modulo-translation augmentation, and a per-modality loss split—and it did not improve on the Cartesian model, performing worse in our runs. We therefore present this as a limitation of our current implementation rather than a general claim about periodic generative geometry; isolating the two components, in particular pairing torus coordinates with an orientation-free but continuous lattice representation, is the most direct next step.

5 Data and Code Availability

The 180 test structures are obtained from the Materials Project database [39]. Their mp-ids are available from our Github repository https://github.com/usccolumbia/cspbenchmark. The code for calculating ranking scores can be also downloaded from the Github repository. The performance metrics calculation is done using the code from the CSPMetrics repository https://github.com/usccolumbia/CSPBenchMetrics.

6 Contribution

Conceptualization, J.H.; methodology,J.H. L. W., S. O., R.D., N.F., Y.S.,E.S., M.X. ; software, L.W., S.O.,Y.S.; resources, J.H.; writing, original draft preparation, J.H., L.W., S.O., N.F., R.D.,E.S.,M.X.; writing and review and editing, J.H., R.D., S.O., N.F.; visualization, L.W.,S.O.; supervision, J.H.; funding acquisition, J.H.

Acknowledgement

We would like to thank the helpful discussion and suggestions of Prof. Yanchao Wang of Jilin University. The research reported in this work was supported in part by National Science Foundation under the grant and 2110033, 2311202, and 2320292. The views, perspectives, and content do not necessarily represent the official views of the NSF.

References

[1] Artem R. Oganov and Colin W. Glass. Crystal structure prediction using ab initio evolutionary techniques: Principles and applications. *The Journal of Chemical Physics*, 124:244704, 2006.

[2] Artem R. Oganov, Yanming Ma, Andriy O. Lyakhov, Mario Valle, and Carlo Gatti. Evolutionary crystal structure prediction and novel high-pressure phases. *Reviews in Mineralogy and Geochemistry*, 71:271–298, 2010.

[3] Artem R Oganov, Chris J Pickard, Qiang Zhu, and Richard J Needs. Structure prediction drives materials discovery. *Nature Reviews Materials*, 4(5):331–348, 2019.

[4] Anubhav Jain, Yongwoo Shin, and Kristin A Persson. Computational predictions of energy materials using density functional theory. *Nature Reviews Materials*, 1(1):15004, 2016.

[5] Richard J Needs and Chris J Pickard. Perspective: Role of structure prediction in materials discovery and design. *APL Materials*, 4(5):053210, 2016.

[6] Yanchao Wang, Jian Lv, Pengyue Gao, and Yanming Ma. Crystal structure prediction via efficient sampling of the potential energy surface. *Accounts of Chemical Research*, 55(15):2068–2076, 2022.

[7] Benjamin J Wieder, Barry Bradlyn, Jennifer Cano, Zhijun Wang, Maia G Vergniory, Luis Elcoro, Mois I Aroyo, and B Andrei Bernevig. Topological materials discovery from crystal symmetry. *Nature Reviews Materials*, 6(9):196–216, 2021.

[8] Maya M Martirossyan, Thomas Egg, Philipp Hoellmer, George Karypis, Mark K Transtrum, Adrian Roitberg, et al. All that structure matches does not glitter. *arXiv preprint arXiv:2509.12178*, 2025.

[9] Scott M. Woodley and Richard Catlow. Crystal structure prediction from first principles. *Nature Materials*, 7:937–946, 2008.

[10] Chris J. Pickard and Richard J. Needs. Ab initio random structure searching. *Journal of Physics: Condensed Matter*, 23:053201, 2011.

[11] Yanchao Wang, Jian Lv, Li Zhu, and Yanming Ma. Crystal structure prediction via particle-swarm optimization. *Physical Review B*, 82:094116, 2010.

[12] Chris J. Pickard and R. J. Needs. Ab initio random structure searching. *Journal of Physics: Condensed Matter*, 23(5):053201, 2011.

[13] Artem R. Oganov and Colin W. Glass. Crystal structure prediction using ab initio evolutionary techniques: Principles and applications. *The Journal of Chemical Physics*, 124(24):244704, 2006.

[14] Yanchao Wang, Jian Lv, Li Zhu, and Yanming Ma. Crystal structure prediction via particle-swarm optimization. *Physical Review B*, 82(9):094116, 2010.

[15] Tian Xie, Xiang Fu, Octavian-Eugen Ganea, Regina Barzilay, and Tommi Jaakkola. Crystal diffusion variational autoencoder for periodic material generation. In *International Conference on Learning Representations (ICLR)*, 2022.

[16] Rui Jiao, Wenbing Huang, Peijia Lin, Jiaqi Han, Pin Chen, Yutong Lu, and Yang Liu. Crystal structure prediction by joint equivariant diffusion. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[17] Rui Jiao, Wenbing Huang, Yu Liu, Deli Zhao, and Yang Liu. Space group constrained crystal generation. In *International Conference on Learning Representations (ICLR)*, 2024.

[18] Zhendong Cao, Xiaoshan Luo, Jian Lv, and Lei Wang. Space group informed transformer for crystalline materials generation. *arXiv preprint arXiv:2403.15734*, 2024. Published in Science Bulletin (2025).

[19] Daniel Levy, Siba Smarak Panigrahi, Sékou-Oumar Kaba, Qiang Zhu, Kin Long Kelvin Lee, Mikhail Galkin, Santiago Miret, and Siamak Ravanbakhsh. Symmcd: Symmetry-preserving crystal generation with diffusion models. *arXiv preprint arXiv:2502.03638*, 2025.

[20] Filip Ekström Kelvinius, Oskar B. Andersson, Abhijith S. Parackal, Dong Qian, Rickard Armiento, and Fredrik Lindsten. Wyckoffdiff: A generative diffusion model for crystal symmetry. *arXiv preprint arXiv:2502.06485*, 2025.

[21] Claudio Zeni, Robert Pinsler, Daniel Zügner, et al. A generative model for inorganic materials design. *Nature*, 2025. DOI: 10.1038/s41586-025-08628-5.

[22] Luis M. Antunes, Keith T. Butler, and Ricardo Grau-Crespo. Crystal structure generation with autoregressive large language modeling. *Nature Communications*, 15(1):10570, 2024.

[23] Benjamin Kurt Miller, Ricky T. Q. Chen, Anuroop Sriram, and Brandon M. Wood. Flowmm: Generating materials with riemannian flow matching. *arXiv preprint arXiv:2406.04713*, 2024.

[24] Xiaoshan Luo, Zhenyu Wang, Qingchang Wang, Xuechen Shao, Jian Lv, Lei Wang, Yanchao Wang, and Yanming Ma. Crystalflow: a flow-based generative model for crystalline materials. *Nature Communications*, 16(1):9267, 2025.

[25] Zhengyang Geng, Mingyang Deng, Xingjian Bai, J. Zico Kolter, and Kaiming He. Mean flows for one-step generative modeling. *arXiv preprint arXiv:2505.13447*, 2025.

[26] Yiyang Lu, Susie Lu, Qiao Sun, Hanhong Zhao, Zhicheng Jiang, Xianbang Wang, Tianhong Li, Zhengyang Geng, and Kaiming He. One-step latent-free image generation with pixel mean flows. *arXiv preprint arXiv:2601.22158*, 2026.

[27] Lai Wei, Sadman Sadeed Omee, Rongzhi Dong, Nihang Fu, Yuqi Song, Edirisuriya Siriwardane, Meiling Xu, Chris Wolverton, and Jianjun Hu. Cspbench: a benchmark and critical evaluation of crystal structure prediction. *arXiv preprint arXiv:2407.00733*, 2024.

[28] Kiyoung Seong, Sungsoo Ahn, Sehui Han, and Changyoung Park. Multimodal crystal flow: Any-to-any modality generation for unified crystal modeling. *arXiv preprint arXiv:2602.20210*, 2026.

[29] Shyue Ping Ong, William Davidson Richards, Anubhav Jain, Geoffroy Hautier, Michael Kocher, Shreyas Cholia, Dan Gunter, Vincent L. Chevrier, Kristin A. Persson, and Gerbrand Ceder. Python materials genomics (pymatgen): A robust, open-source python library for materials analysis. *Computational Materials Science*, 68:314–319, 2013.

[30] Atsushi Togo, Kohei Shinohara, and Isao Tanaka. Spglib: a software library for crystal symmetry search. *Science and Technology of Advanced Materials: Methods*, 4(1):2384822, 2024.

[31] Anubhav Jain, Shyue Ping Ong, Geoffroy Hautier, Wei Chen, William Davidson Richards, Stephen Dacek, Shreyas Cholia, Dan Gunter, David Skinner, Gerbrand Ceder, and Kristin A. Persson. Commentary: The materials project: A materials genome approach to accelerating materials innovation. *APL Materials*, 1(1):011002, 2013.

[32] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. Pytorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[33] Lai Wei, Rongzhi Dong, Nihang Fu, Sadman Sadeed Omee, and Jianjun Hu. Tcsp 2.0: Template based crystal structure prediction with improved oxidation state prediction and chemistry heuristics. *arXiv preprint arXiv:2503.23183*, 2025.

[34] Leopold Talirz, Snehal Kumbhar, Elsa Passaro, Aliaksandr V Yakutovich, Valeria Granata, Fernando Gargiulo, Marco Borelli, Martin Uhrin, Sebastiaan P Huber, Spyros Zoupanos, et al. Materials cloud, a platform for open computational science. *Scientific Data*, 7(1):299, 2020.

[35] Morten Niklas Gjerding, Alireza Taghizadeh, Asbjørn Rasmussen, Sajid Ali, Fabian Bertoldo, Thorsten Deilmann, Nikolaj Rørbæk Knøsgaard, Mads Kruse, Ask Hjorth Larsen, Simone Manti, et al. Recent progress of the computational 2d materials database (c2db). *2D Materials*, 8(4):044002, 2021.

[36] Amil Merchant, Simon Batzner, Samuel S Schoenholz, Muratahan Aykol, Gowoon Cheon, and Ekin Dogus Cubuk. Scaling deep learning for materials discovery. *Nature*, 624(7990):80–85, 2023.

[37] Benjamin Rhodes, Sander Vandenhaute, Vaidotas Šimkus, James Gin, Jonathan Godwin, Tim Duignan, and Mark Neumann. Orb-v3: atomistic simulation at scale. *arXiv preprint arXiv:2504.06231*, 2025.

[38] Bowen Deng, Peichen Zhong, KyuJung Jun, Janosh Riebesell, Kevin Han, Christopher J. Bartel, and Gerbrand Ceder. CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling. *Nature Machine Intelligence*, 5(9):1031–1041, 2023.

[39] Anubhav Jain, Shyue Ping Ong, Geoffroy Hautier, Wei Chen, William Davidson Richards, Stephen Dacek, Shreyas Cholia, Dan Gunter, David Skinner, Gerbrand Ceder, et al. Commentary: The materials project: A materials genome approach to accelerating materials innovation. *APL materials*, 1(1), 2013.